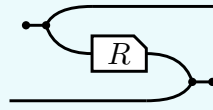


Relation Algebra

§3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure **DRAWS** that — the picture and the formula below are R° , not its definition:



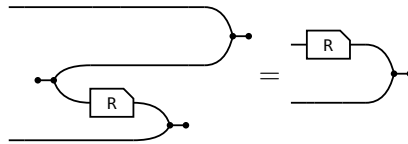
$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where $\circ \triangleleft : \mathbb{I} \rightarrow \mathbf{a} \otimes \mathbf{a}$ opens a pair of wires out of nothing and $\triangleright \circ : \mathbf{b} \otimes \mathbf{b} \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\circ^\circ$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



(3.1a)

R° is **DEFINED** as the bending of $(R \otimes 1) \triangleright \circ$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circ \triangleleft$ bends the $\mathbf{a}$ strand down and the $\triangleright \circ$ brings it back up, while the $\mathbf{b}$ strand rides through untouched. Nothing else is spent below.